Please circle your Class Section: C01 (Dr. Davidson) C02 (Dr. Athar)

Name: ______________________ Student ID: ______________________

ELECENG 3CL4 (Introduction to Control Systems)
Midterm Examination

Department of Electrical and Computer Engineering, McMaster University
Instructors: Dr. Tim Davidson (C01) and Dr. Shahrukh Athar (C02), Winter 2025
Thursday, March 6, 2025, 6:45 PM to 8:45 PM (120 minutes)

Examination Conditions This test is conducted under standard examination conditions. In particular:

- You are not to communicate in any manner with any other student.
- You are not to look at the questions nor start writing until told to do so by an invigilator.
- When the invigilator announces that the test is over, you are to stop writing *immediately*.

The penalties for violating these or any other standard examination conditions will be severe, including a zero on the test and the commencement of formal disciplinary proceedings.

Instructions

- There are five (5) questions on this examination paper. Answer all five questions in the space provided in this exam paper. You may use the back of the pages if necessary.
- **When given permission to start writing, you must ensure that your copy of the exam paper is complete and bring any discrepancies to the attention of your exam invigilator.**
- **Write your name and student number (nine-digit McMaster ID)** on the front page of this examination paper. **Encircle your section** at the top of this page.
- Write your answers neatly and in a logical fashion. We may refuse to mark answers which are difficult to decipher.
- Direct answers without proper justifications will lead to a reduced grade. However, justifications should be concise and to-the-point.
- You may write your test in pencil, but booklets written in pencil or with pages torn out will not be re-graded.
- **Closed-books, closed-notes.**
- **Aids permitted:**

  McMaster Standard Calculator (Casio FX-991MS/MS+).
- Total Marks: 100 (mark allocations for questions/parts are shown within questions).

Q1 (20)	Q2 (20)	Q3 (20)	Q4 (20)	Q5 (20)	Total (100)

Question 1 (20 marks) This question is composed of five short parts (a-e). **Answers are expected to be short and concise**. We have provided more space than is necessary. Answer the following questions:

a) (4 marks) In the figure given below, the input to a linear process is the unit step function, i.e., $x(t) = u(t)$, and the output for $t \geq 0$ is $y(t) = 1 - 3e^{-4t} + 2e^{-6t}$.

$$x(t) \longrightarrow \boxed{\text{Process}} \longrightarrow y(t)$$

Determine the transfer function of the process.

$$x(t) = u(t) \longleftrightarrow \frac{1}{s} = X(s)$$

$$y(t) = 1 - 3e^{-4t} + 2e^{-6t} \longleftrightarrow \frac{1}{s} - 3\frac{1}{s+4} + 2\frac{1}{s+6} = Y(s)$$

$$\frac{1}{s} - \frac{3}{s+4} + \frac{2}{s+6}$$

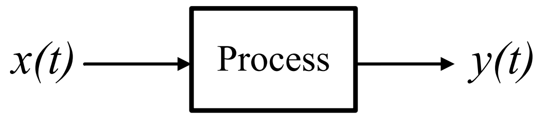
$$\frac{1(s+4)(s+6) - 3(s+6) + 2(s+4)}{s(s+4)(s+6)}$$

$$T(s) = \frac{24}{(s+4)(s+6)}$$

$$T(s) = \frac{Y(s)}{X(s)} = \frac{\frac{1}{s} - 3\frac{1}{s+4} + 2\frac{1}{s+6}}{1/s} = 1 - 3s\frac{1}{s+4} + 2s\frac{1}{s+6}$$

Question 1 (20 marks) This question is composed of five short parts (a-e). **Answers are expected to be short and concise**. We have provided more space than is necessary. Answer the following questions:

a) (4 marks) In the figure given below, the input to a linear process is the unit step function, i.e., $x(t) = u(t)$, and the output for $t \geq 0$ is $y(t) = 1 - 3e^{-4t} + 2e^{-6t}$.

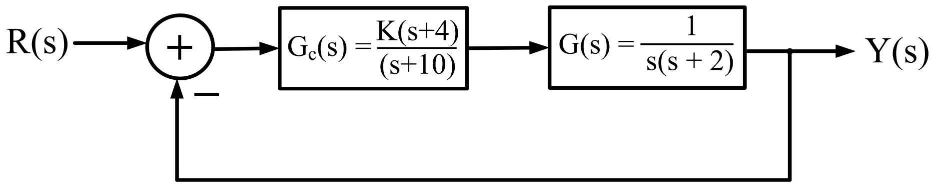


Determine the transfer function of the process.

✗ no attempt 2

ramp input : $x(t) = At \longrightarrow \frac{A}{s^2} = X(s)$

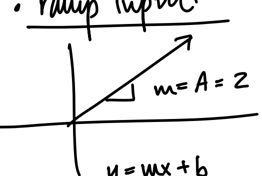
b) **(4 marks)** Consider the closed-loop feedback control system given below.



Determine the condition on the controller parameter K to have a steady-state error of less than 12.5% for a ramp input with slope $A = 2$.

- $K = ?$
- $e_{ss} = 12.5\%$

- ramp input.



$y = mx + b$
$y = 2x$
$x(t) = 2t$

$x(t) = 2t \longrightarrow \frac{2}{s^2} = X(s)$

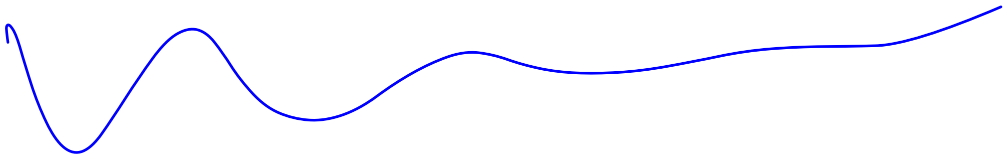
- 1p @ origin : Type 1
- ramp input & $T_d(s) = 0$

$e_{ss} = \frac{A}{k_v} \longrightarrow \quad K_v = \frac{A}{e_{ss}} = \frac{2}{0.125} =$

- formula sheet: $K_v = \lim_{s \to 0} s\, G_c(s) G(s)$

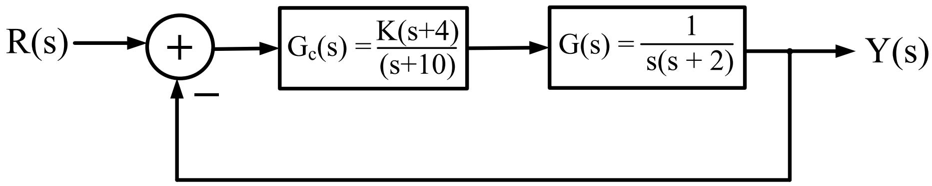
$$= \cancel{\lim_{s \to 0}} \cancel{s} \cdot \frac{k(s+4)}{s+10} \cdot \frac{1}{\cancel{s}(s+2)}$$

$$\frac{2}{0.125} = \frac{k(s+4)}{(s+10)(s+2)}$$



* attempt #2

b) (4 marks) Consider the closed-loop feedback control system given below.



Determine the condition on the controller parameter K to have a steady-state error of less than 12.5% for a ramp input with slope $A = 2$.

- K condition?
- $e_{ss} \leq 12.5\%$
- ramp input: $R(s) = \frac{A}{s^2} = \frac{2}{s^2}$

- $G_c(s)\,G(s) \rightarrow$ 1 pole @ origin
- ramp input
- no $T_d(s)$

Type 1 & ramp input: $e_{ss} = \frac{A}{K_v}$

formula sheet: $K_v = \lim_{s \to 0} s\, G_c(s)\, G(s)$

$= \lim_{s \to 0} s \frac{K(s+4)}{\cancel{s}(s+2)(s+10)}$

$= \cancel{\lim_{s \to 0}} \frac{\cancel{Ks} + K4}{\cancel{s^2} + \cancel{12s} + 20}$

$= \frac{4K}{20}$

$= \frac{K}{5}$

$e_{ss} = \frac{A}{K_v} \rightarrow e_{ss} = 12.5\%\, A$

$= 0.125\, A$

$\downarrow$

$0.125\cancel{A} > \frac{\cancel{A}}{K_v}$

$0.125 > \frac{1}{K_v}$

$0.125 > \frac{1}{K/5}$

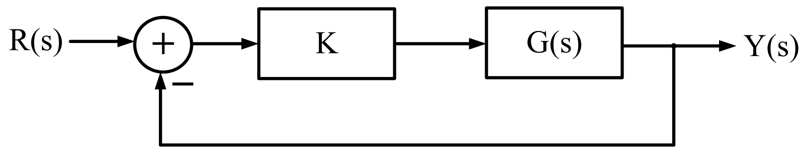
$> \frac{5}{K}$

$K 0.125 > 5$

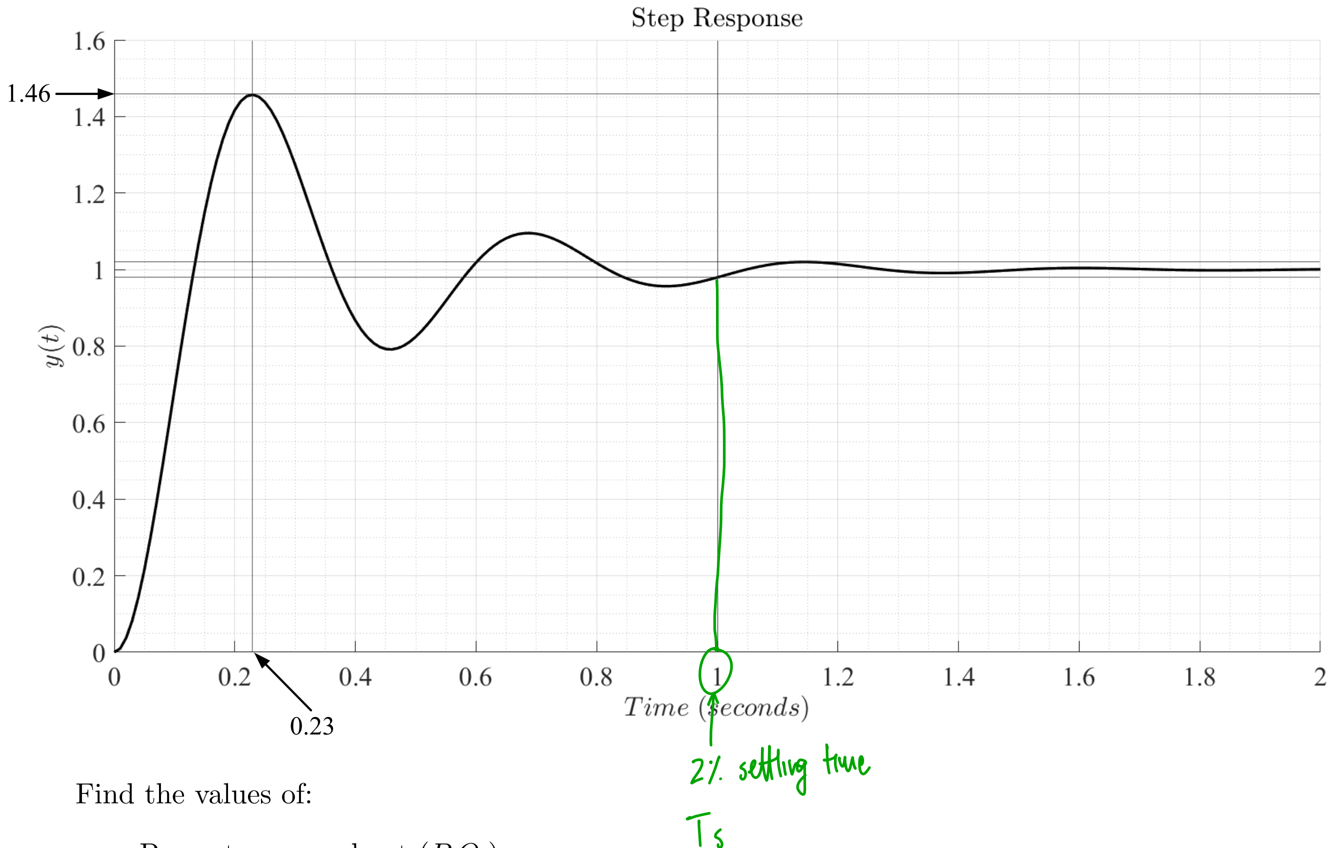
$K > \frac{5}{0.125}$

$\therefore K > 40$

c) (5 marks) Consider the closed-loop feedback control system given below where $G(s)$ is a second-order process.



A unit step input is given to this system, i.e., $r(t) = u(t)$. The plot of the step response, $y(t)$, is shown in the figure below where a ±2% corridor around the height of input step has been marked.



Find the values of:

- Percentage overshoot ($P.O.$)
- 2% settling time (T_s)
- Peak time (T_p)
- Steady-state value (fv)
- Steady-state error (e_{ss})

- $P.O. = \frac{1.46 - 1}{1} = 0.46 = 46\%$
- 2% settling time : within corridor : $T_s = 1s$
- $T_p = 0.23s$
- $e_{ss} = 0$
- $f_v = 1$

d) (5 marks) For the control system and step response plot of part c), do the following:

- How many closed-loop poles are there?
- Find the location of the closed-loop poles in the *s-plane* and provide a sketch that depicts them in the *s-plane*. (Note: Exact pole locations are required).

∘ 2nd order sys. ⟶ 2 poles

∘ underdamped → 2 conj pair on LHP

↳ formula sheet: underdamped → $0 < \zeta < 1$: poles $= -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$

$$\zeta = \frac{-\ln(PO./100)}{\sqrt{\pi^2 + \ln(PO/100)}}$$

$$=$$

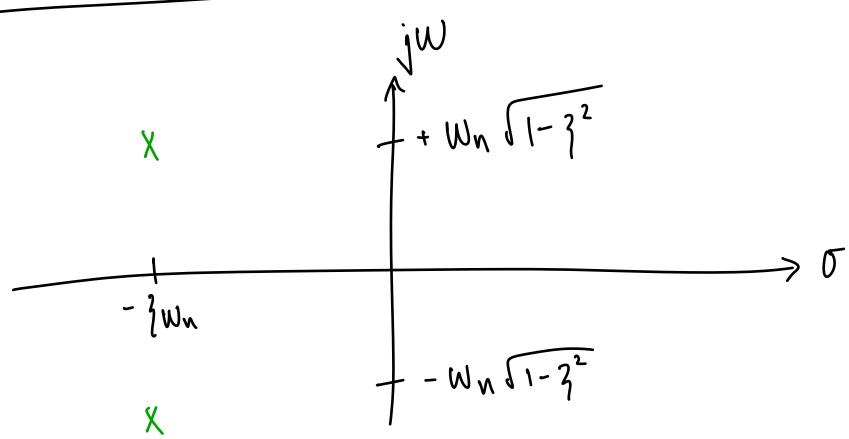
$$= 0.24$$

$$T_s = \frac{4}{\zeta\omega_n}$$

$$\omega_n = \frac{4}{\zeta T_s}$$

$$=$$

pole sketch:



e) (2 marks) For the control system and step response plot of part c), consider the following statement:

- *The type-number of the given system is Type-0.*

State whether the above statement is true or false and justify your answer. (Note: There are no marks for just stating true or false, proper justification is required to earn marks).

false.

for Type 0: $e_{ss} = 0$

this case: $e_{ss} \neq 0$

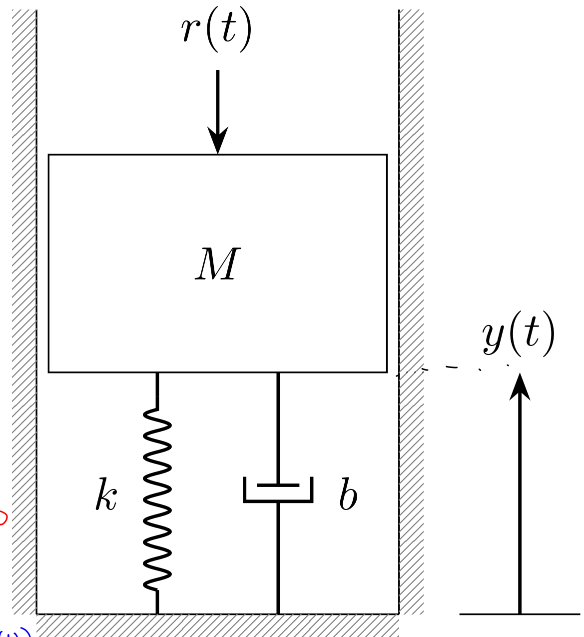
Question 2 (20 marks) Consider the following mass-spring-damper system, which is arranged vertically. The mass is balanced on top of the spring and damper. The side walls have negligible friction.

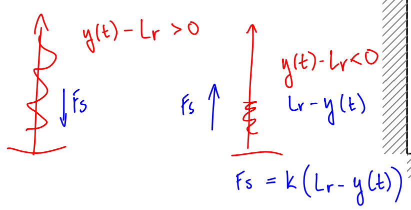
$* \; F_s = k(L_r - y(t))$

$= k(\Delta x)$

$* \; +\Delta x =$ stretch

$-\Delta x =$ compress





a) (10 marks) Assume that for $t < 0$ the applied force $r(t)$ in Newtons (N) has been zero for long enough for the mass to have reached its equilibrium position. Derive a differential equation that describes the position of the bottom surface of the mass, $y(t)$, in metres (m), for time $t \geq 0$ in seconds (s).

(Hint: Among other things, this equation will be a function of the mass M, in kilograms, the spring constant k, in Nm^{-1}, the relaxed length of the spring, L_r, in m, the damper coefficient b, in Nsm^{-1}, and the acceleration due to gravity, g, in ms^{-2}.)

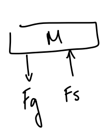
b) (6 marks) Derive the transfer function from $R(s)$ to $Y(s)$

c) (4 marks) If the applied force $r(t)$ is a step function of size r_0 N, what is the final position of the bottom surface of the mass.

a)

• FBD:

$t < 0$:



• Equilibrium:

$\Sigma F = \cancel{ma} = 0$

$F_g = F_s$

$Mg = k(L_r - y_i)$

$Mg = kL_r - ky_i$

$ky_i = \frac{kL_r - Mg}{k}$

$y_i = L_r - \frac{1}{k}Mg$

• $t \geq 0$:

$\Sigma F = ma$

• $F_g = Mg$

• $F_d = -by'(t)$

• $F_s = k(L_r - y(t))$

$My''(t) = F_s + F_d - r(t) - F_g$

$= k(L_r - y(t)) - by'(t) - r(t) - Mg$

$My''(t) = kL_r - ky(t) - by'(t) - r(t) - Mg$

$\therefore \; My''(t) + by'(t) + ky(t) = +kL_r - r(t) - Mg$

$\downarrow \mathcal{L}$

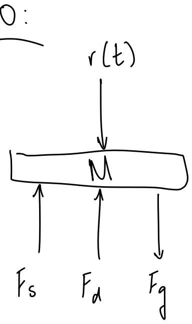
b) $Ms^2Y(s) + bsY(s) + kY(s) = \frac{kL_r}{s} - R(s) - \frac{Mg}{s}$

$Y(s)\left[Ms^2 + bs + k\right] = \frac{kL_r}{s} - \frac{Mg}{s} - R(s)$

$Y(s) = \frac{-1}{Ms^2+bs+k}R(s) + \frac{1}{Ms^2+bs+k}\frac{kL_r}{s} - \frac{1}{Ms^2+bs+k}\frac{Mg}{s}$

$\therefore \; T(s) = \frac{Y(s)}{R(s)} = \frac{-1}{Ms^2+bs+k}$ → rest inputs = 0

$t \geq 0$:



[More space for your answer to Question 2.]

c) input: step func. $A = r_0$

$$R(s) = \frac{A}{s} = \frac{r_0}{s}$$

$$y_{ss} = \lim_{t \to \infty} y(t)$$

$$= \lim_{s \to 0} s\, Y(s)$$

$$= \lim_{s \to 0} s \cdot \left[\frac{1}{Ms^2 + bs + k} \left(-R(s) + \frac{kL_r}{s} - \frac{Mg}{s} \right) \right]$$

$$= \lim_{s \to 0} s \cdot \left[\frac{1}{Ms^2 + bs + k} \left(\frac{-r_0}{s} + \frac{kL_r}{s} - \frac{Mg}{s} \right) \right]$$

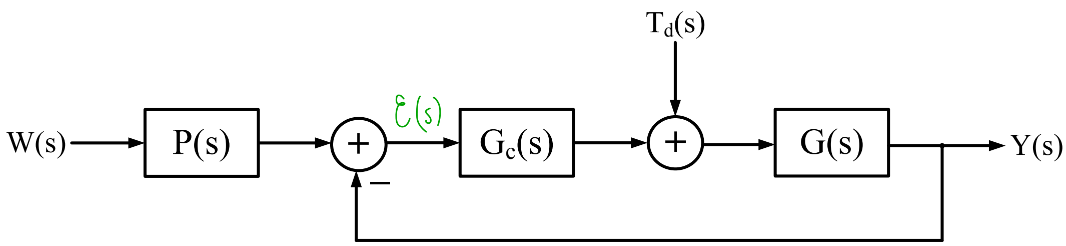
$$= \lim_{s \to 0} \cancel{s} \cdot \left[\frac{1}{Ms^2 + bs + k} \frac{1}{\cancel{s}} \cdot (-r_0 + kL_r - Mg) \right]$$

$$= \cancel{\lim_{s \to 0}} \frac{1}{\cancel{Ms^2} + \cancel{bs} + k} (-r_0 + kL_r - Mg)$$

$$= \frac{1}{k} (-r_0 + kL_r - Mg)$$

$$\therefore y_{ss} = \frac{-r_0}{k} + L_r - \frac{Mg}{k}$$

Question 3 (20 marks) Consider the following block diagram



a) **(5 marks)** For the case where $T_d(s) = 0$, derive the transfer function from $W(s)$ to $Y(s)$.

b) **(3 marks)** For the case where $P(s) = \frac{4}{s+4}$, $G(s) = \frac{1}{s+3}$, and $G_c(s) = K$, write the transfer function in part a) as the ratio of two polynomials in s. $\longrightarrow$ means simplify $\to$ no fractions in num & denom.

c) **(6 marks)** Find a value of K for which the steady-state error due to a step input $w(t)$ (when $T_d(s) = 0$) is less than 3% of the size of the step.

d) **(3 marks)** Design a controller $G_c(s)$ such that the steady-state error due to a step input is zero (when $T_d(s) = 0$).

e) **(3 marks)** How do the transfer functions in parts a) and b) change when $T_d(s) \neq 0$?

b) $T(s) = \dfrac{4k}{s^2 + (k+7)s + 4(k+3)}$

c)
- Step input: $A \to W(s) = \frac{A}{s}$
- $e_{ss} < 0.03A$
- $k = ?$

- Step input:
- no poles @ origin
- $T_d(s) = 0$

$\to$ Type 0, $e_{ss} = \dfrac{A}{1+K_p}$

- $K_{posn} = \lim_{s\to 0} L(s)$

$= \cancel{\lim_{s\to 0}} \dfrac{4k}{\cancel{s^2} + \cancel{(k+7)s} + 4(k+3)}$

$= \dfrac{\cancel{4}k}{\cancel{4}(k+3)}$

$= \dfrac{k}{k+3} \begin{matrix}\times 1/k \\ \times 1/k\end{matrix}$

$= \dfrac{1}{1+3/k}$

calc error of K_{posn}

$e_{ss} = \dfrac{A}{1+K_p}$

$0.03\cancel{A} > \dfrac{\cancel{A}}{1+K_p}$

$0.03 > \dfrac{1}{1+K_p}$

$1+K_p > \dfrac{1}{3/100}$

$1+K_p > \dfrac{100}{3}$

$K_p > \dfrac{100}{3} - 1$

$\dfrac{1}{\cancel{1+3/k}} > \dfrac{97}{3}(1+3/k)$

$\dfrac{3}{97} > 1 + 3/k$

$\dfrac{3}{97} - 1 > \dfrac{3}{k}$

$k > \dfrac{3}{\frac{3}{97} - 1}$

$k > -3.1$

↳ oops... calc error

- $K_{posn} = \lim_{s\to 0} G_c(s)G(s)$

$= \cancel{\lim_{s\to 0}} K \cdot \dfrac{1}{\cancel{s}+3}$

$= \dfrac{k}{3}$

correct value for K_{posn}

$k > 97$ = correct answer

- $P(s) = 1 \to$ DC gain
  ↳ doesnt contribute to error
  ↳ $k > 97$ is enough.

[More space for your answer to Question 3.]

d) ∘ $G_c(s) = ?$

∘ $e_{ss} = 0$

∘ $T_d(s) = 0$

∘ step input

→ need at least Type 1

↳ 1 pole @ origin

$G_c(s)$ = PI controller

$= k_p + \frac{1}{s} k_i = \frac{k_p s + k_i}{s}$

∘ $G_c(s)G(s)$

$= \frac{1}{s+3} \cdot \frac{k_p s + k_i}{s}$

$= \frac{k_p s + k_i}{s(s+3)}$

* $k_i > 0$ & check for stable loop

↳ check loop stability:

$$\frac{G_c(s)\,G(s)}{1 + G_c(s)\,G(s)} = \frac{\frac{k_p s + k_i}{s(s+3)}}{1 + \frac{k_p s + k_i}{s(s+3)}} = \frac{\sim\sim}{\frac{s^2 + 3s + k_p s + k_i}{s(s+3)}} = \frac{k_p s + k_i}{\cancel{s(s+3)}} \cdot \frac{\cancel{s(s+3)}}{s^2 + s(3+k_p) + k_i}$$

$$= \frac{k_p s + k_i}{s^2 + s(3+k_p) + k_i}$$

↳ need $3 + k_p > 0$

$k_p > -3$ & $k_i > 0$

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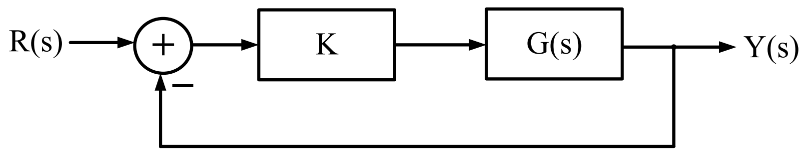
e) it doesn't.

$$Y(s) = T(s)W(s) + F(s)T_d(s)$$

intended input

not ripple input

Question 4 (20 marks) Consider the following block diagram.



a) (3 marks) Show that when $G(s) = \frac{1}{s(s+4)}$ the closed-loop poles are at positions $s = -2 \pm \sqrt{4-K}$.

b) (5 marks) If $G(s) = \frac{1}{s(s+4)}$, where are the poles if the response $y(t)$ to a step input $r(t)$ has 20% overshoot?

c) (4 marks) What is the value of K that puts the poles in the positions in part b)?

d) (4 marks) What are the corresponding values for the peak time and the 2% settling time?

e) (4 marks) Describe, both qualitatively and quantitatively what happens to the percentage overshoot, the peak time and the 2% settling time when the value of K in part c) is doubled.

a) $$T(s) = \frac{kG(s)}{1+kG(s)} = \frac{k\frac{1}{s(s+4)}}{1+k\frac{1}{s(s+4)}} \begin{matrix} \times\, s(s+4) \\ \times\, s(s+4) \end{matrix} = \frac{k}{s(s+4)+k}$$

$$= \frac{k}{s^2+4s+k}$$

closed loop poles: $0 = s^2 + 4s + k$

$$s = \frac{-4 \pm \sqrt{4^2 - 4(1)k}}{2(1)}$$

$$= \frac{-4 \pm \sqrt{16-4k}}{2}$$

$$= \frac{-4 \pm \sqrt{4(4-k)}}{2}$$

$$= \frac{-4 \pm 2\sqrt{4-k}}{2}$$

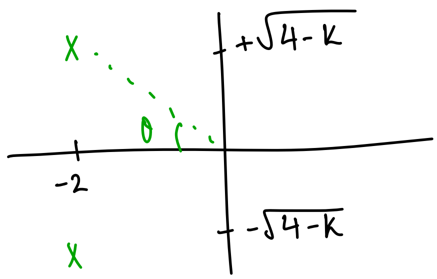
$$= -2 \pm \sqrt{4-k}$$

[More space for your answer to Question 4.]

b) $r(t)$ = Step input

P.O. = 20%

$$\zeta = \frac{-\ln(PO/100)}{\sqrt{\pi^2 + \ln^2(PO/100)}} \doteq 0.456$$



$$T_s = \frac{4}{\zeta \omega_n} \Big] \text{ real part}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} \Big] \text{ img part}$$

real: $-2 = \zeta \omega_n$

img:

method 2:

$\theta = \cos^{-1}(\zeta)$

$\doteq 62.87°$

$\tan\theta = \frac{opp}{adj}$

$y = 2\tan\theta$

$= 3.9$

$s_{1,2} = -2 \pm j\,3.9$

c) $k = ?$

$s_{1,2} = -2 \pm j\sqrt{k-4}$

$= -2 \pm j(3.9)$

$3.9 = \sqrt{k-4}$

$3.9^2 = k-4$

$k = 3.9^2 + 4$

⋮

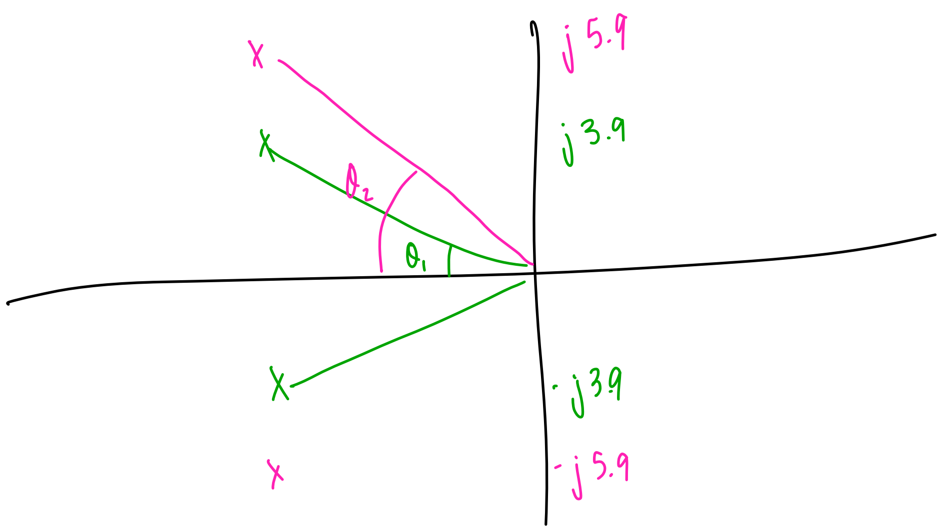
$k = 19.21$

[More space for your answer to Question 4.]

d) $T_s = ?$

$T_p = ?$] use formula sheet

e) $PO = ?$ $T_s = ?$ $T_p = ?$ when $k = ?$



$T_s = \frac{4}{real} = 2s$

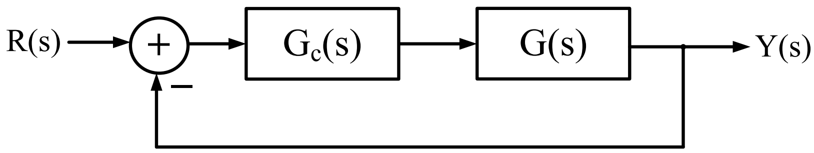
$T_p = \frac{\pi}{img} = \frac{\pi}{5.9}$

$\tan\theta = \frac{opp}{adj}$

$\theta = \tan^{-1}\left(\frac{opp}{adj}\right)$

- qualitative:
  - double k:
    - PO ↑
    - Tp ↓
    - 2% time constant

Question 5 (20 marks)



Consider the closed-loop feedback control system above where $G_c(s) = \frac{K(s+1)}{(s+7)}$ and $G(s) = \frac{1}{s(s-2)}$. ← $s = +2$ ↳ unstable

a) **(2 marks)** Find the open-loop transfer function L(s). What is the type number of the system?

b) **(4 marks)** Find the closed-loop transfer function, $T(s) = \frac{Y(s)}{R(s)}$.

c) **(6 marks)** Identify the characteristic equation and construct the complete Routh-Hurwitz table for this system.

d) **(3 marks)** Find the range of K for which the system is stable.

e) **(5 marks)** What value of K puts a pair of closed-loop poles on the $j\omega$-axis, what is the position of the $j\omega$-axis poles?

a) $L(s) = G_c(s)G(s) = \frac{k(s+1)}{s(s+7)(s-2)}$, Type 1 → 1 pole @ origin

b) $$T(s) = \frac{G_c(s)G(s)}{1+G_c(s)G(s)} = \frac{L(s)}{1+L(s)} = \frac{\frac{k(s+1)}{s(s+7)(s-2)} \times s(s+7)(s-2)}{1+\frac{k(s+1)}{s(s+7)(s-2)} \times s(s+7)(s-2)}$$

$$= \frac{k(s+1)}{s(s+7)(s-2)+k(s+1)}$$

$$= \frac{k(s+1)}{s(s^2+5s-14)+ks+k}$$

$$= \frac{k(s+1)}{s^3+5s^2-14s+ks+k}$$

$$= \frac{ks+1}{s^3+5s^2+(k-14)s+k}$$

$$0 = s^3+5s^2+(k-14)s+k$$

[More space for your answer to Question 5.]

- Char. Eq: $0 = s^3 + 5s^2 + (k-14)s + k$
- R.H. table: 3rd order → $3+1=4$ rows

s^3	row	3	1	$k-14$	0
s^2	⋮	2	5	k	0
s^1	⋮	1	b_1	b_0	
s^0	⋮	0	c_1		

$$b_1 = \frac{-1}{5}\begin{vmatrix} 1 & k-14 \\ 5 & k \end{vmatrix}$$
$$= \frac{-1}{5}\left(k - 5(k-14)\right)$$
$$= \frac{-1}{5}\left(k - 5k + 70\right)$$
$$= \frac{-1}{5}(k)(1-5+70) = \frac{-k}{5}(74)$$

$$b_0 = \frac{-1}{5}\begin{vmatrix} \# & 0 \\ \# & 0 \end{vmatrix}$$
$$= 0$$

$$c_1 = \frac{-1}{b_1}\begin{vmatrix} 5 & k \\ b_1 & b_0 \end{vmatrix}$$
$$= \frac{\cancel{-}1}{\cancel{b_1}}\left(\cancel{5(0)} + \cancel{b_1}k\right)$$
$$= k$$

↓

s^3	row	3	1	$k-14$	0
s^2	⋮	2	5	k	0
s^1	⋮	1	$\frac{4k-70}{5}$	0	
s^0	⋮	0	K		

↓ no sign change

- row 1: $\frac{4k-70}{5} > 0$

$4k - 70 > 0$

$4k > 70$

$k > 70/4$

> 17.5

- row 0: $k > 0$

$k > 0 > 17.5$

$\boxed{\therefore k > 17.5 \text{ for stability}}$

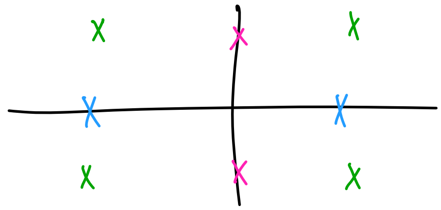
[More space for your answer to Question 5.]

e) • $k = ?$ for poles on $j\omega$-axis.

• position of poles

↳ stable @ $k > 17.5$

↳ @ $j\omega$ axis : $k = 17.5$



radially symmetry:

1) diagonal → 4 poles @ real & img
2) on σ axis → 2 real
3) on $j\omega$ axis → 2 img.

• need 2 poles → occurs on row 2 → auxillary row = row 2

$a(s) = 5$

→ auxilly row needs full zero row below it.

↳ row 1 = 0's

$$\frac{4k-10}{5} = 0$$

$$k = 17.5$$

→ $a(s) = ?$

End of Examination

Useful Information

- The solutions to the quadratic equation $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- In the function $f(t) = Ae^{-t/\tau}$ the term τ is sometimes referred to as the time constant.
- The following Laplace transform pairs may be of use. Here $u(t)$ is the unit step function

$$
\begin{aligned}
u(t) &\longleftrightarrow \frac{1}{s} \\
e^{-at}u(t) &\longleftrightarrow \frac{1}{s+a} \\
sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{s^2+\omega^2} \\
cos(\omega t)u(t) &\longleftrightarrow \frac{s}{s^2+\omega^2} \\
e^{-at}\cos(\omega t)u(t) &\longleftrightarrow \frac{s+a}{(s+a)^2+\omega^2} \\
e^{-at}\sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{(s+a)^2+\omega^2} \\
t^n u(t) &\longleftrightarrow \frac{n!}{s^{n+1}} \\
\frac{df(t)}{dt} &\longleftrightarrow sF(s) - f(0^-) \\
\frac{d^2f(t)}{dt^2} &\longleftrightarrow s^2F(s) - sf(0^-) - f'(0^-) \\
\int_{0^-}^{t} f(\lambda)\,d\lambda &\longleftrightarrow \frac{F(s)}{s}
\end{aligned}
$$

- A generic form for a rational Laplace Transform is $G(s) = K_G \frac{\prod_{i=1}^{M}(s+z_i)}{\prod_{j=1}^{n}(s+p_j)}$, where the set of finite zeros is $\{-z_i\}_{i=1}^{M}$ and the set of poles is $\{-p_j\}_{j=1}^{n}$.
- The final value theorem states that for a Laplace Transform $Y(s)$ with all its poles in the left half plane, except for at most one pole at the origin, $\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s)$.
- For a standard second-order system with a transfer function of the form $T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$, when $0 < \zeta < 1$, the poles are at positions $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. The step response of such a system has the following properties:
  - $y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin\left(\omega_n\sqrt{1-\zeta^2}\,t + \theta\right)$, where $\theta = \cos^{-1}(\zeta)$.
  - 2% settling time, $T_s \approx \frac{4}{\zeta\omega_n}$.
  - Percentage (maximum) overshoot calculated from step response, $P.O. = 100\left[\frac{M_{pt}-fv}{fv}\right]$, where M_{pt} is the peak value and fv is the steady-state value of the step response.
  - Percentage (maximum) overshoot (given ζ), $P.O. = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$. Hence, $\zeta = \frac{-\ln(P.O./100)}{\sqrt{\pi^2+\ln^2(P.O./100)}}$.
  - Peak time, $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$.
- Steady state error constants:
  - Position error constant: $K_{\text{posn}} = \lim_{s\to 0} G_c(s)G(s)$
  - Velocity error constant: $K_v = \lim_{s\to 0} sG_c(s)G(s)$